

QA LAB BRASIL

Reference guide - QA Lab Brasil

Playwright

Modern E2E web automation with Python and Pytest examples.

What is Playwright?

Playwright is an automation tool designed for end-to-end testing. In Python, the official Pytest plugin provides browser fixtures, context isolation and execution on Chromium, Firefox and WebKit.

Python installation

Create and activate a virtual environment, run `pip install pytest-playwright`, then `playwright install` to download the required browsers. Validate with `pytest`.

First test

A typical test opens a page, locates elements by role, label or text, and uses web-first assertions. Prefer semantic locators over fragile selectors.

Headless and headed

Pytest runs headless by default. Use `pytest --headed` to watch the browser and `--browser` to select engines.

Debug and evidence

Use Inspector, tracing, screenshots and videos to investigate failures. Evidence should support diagnosis, not only prove failure.

CI/CD

Playwright can run in pipelines: install dependencies and browsers, run `pytest` and publish failure traces as artifacts.

Good practices

Isolate data, avoid fixed sleeps, keep assertions explicit, store secrets in environment variables and treat flaky tests as suite defects.

QA Tip: ☐ QA Tip: automate stable, critical and repetitive flows first.

Official: <https://playwright.dev/python/docs/intro>